

3. $|x + 3| + |x - 1| \geq 5$

Si igualam els valors absoluts a 0, treuem dues arrels, tals que: $x = -3$ i $x = 1$

• **Cas 1:** $x < -3$

$$-(x + 3) - (x - 1) \geq 5 \implies -x - 3 - x + 1 \geq 5 \implies -2x \geq 7 \implies x \leq -3.5$$

Treim Sol.1: $x \in (-\infty, -3.5]$

• **Cas 2:** $-3 \leq x < 1$

$$(x + 3) - (x - 1) \geq 5 \implies x + 3 - x + 1 \geq 5 \implies 4 \geq 5 \text{ (Fals)}$$

Treim Sol.2: $\emptyset$

• **Cas 3:** $x \geq 1$

$$(x + 3) + (x - 1) \geq 5 \implies x + 3 + x - 1 \geq 5 \implies 2x \geq 3 \implies x \geq 1.5$$

Treim Sol.3: $[1.5, +\infty)$

Si unim les tres solucions, tenim que: $x \in (-\infty, -3.5] \cup [1.5, +\infty)$